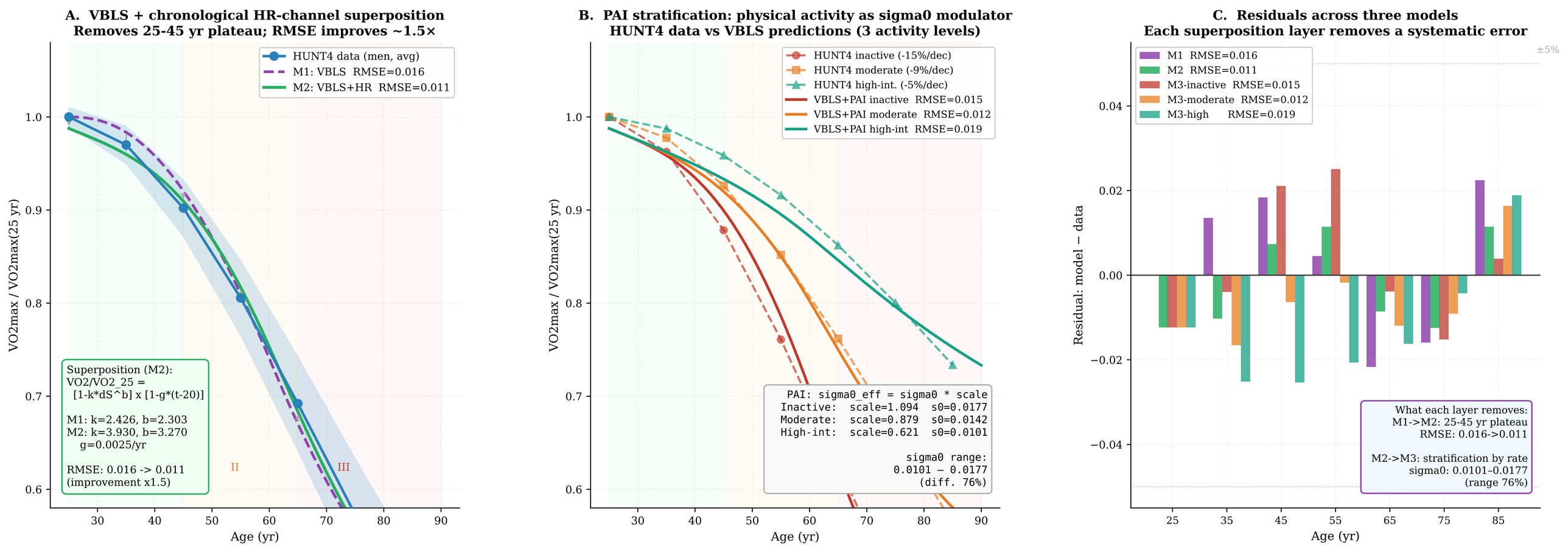

A. VBLS + chronological HR-channel superposition
Removes 25-45 yr plateau; RMSE improves ~1.5×
HUNT4 data (men, avg)
M1: VBLS RMSE=0.016
M2: VBLS+HR RMSE=0.011
Superposition (M2):
VO2/VO2_25 =
[1-k*dS^b] x [1-g*(t-20)]
M1: k=2.426, b=2.303
M2: k=3.930, b=3.270
g=0.0025/yr
RMSE: 0.016 -> 0.011
(improvement x1.5)
II
III
VO2max / VO2max(25 yr)
Age (yr)
B. PAI stratification: physical activity as sigma0 modulator
HUNT4 data vs VBLS predictions (3 activity levels)
HUNT4 inactive (-15%/dec)
HUNT4 moderate (-9%/dec)
HUNT4 high-int. (-5%/dec)
VBLS+PAI inactive RMSE=0.015
VBLS+PAI moderate RMSE=0.012
VBLS+PAI high-int RMSE=0.019
PAI: sigma0_eff = sigma0 * scale
Inactive: scale=1.094 s0=0.0177
Moderate: scale=0.879 s0=0.0142
High-int: scale=0.621 s0=0.0101
sigma0 range:
0.0101 – 0.0177
(diff. 76%)
VO2max / VO2max(25 yr)
Age (yr)
C. Residuals across three models
Each superposition layer removes a systematic error
M1 RMSE=0.016
M2 RMSE=0.011
M3-inactive RMSE=0.015
M3-moderate RMSE=0.012
M3-high RMSE=0.019
±5%
What each layer removes:
M1->M2: 25-45 yr plateau
RMSE: 0.016->0.011
M2->M3: stratification by rate
sigma0: 0.0101–0.0177
(range 76%)
Residual: model − data
Age (yr)